# Polar Curves and Derivatives

ANSWER KEY

## Converting between polar and rectangular coordinates

**1** Convert the polar point $\left(4, \frac{\pi}{3}\right)$ to rectangular coordinates.

$x = r\cos\theta = 4\cos\left(\frac{\pi}{3}\right) = 4\left(\frac{1}{2}\right) = 2$. $y = r\sin\theta = 4\sin\left(\frac{\pi}{3}\right) = 4\left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}$. Point: $(2, 2\sqrt{3})$.

**2** Convert the polar equation $r = 4\cos\theta$ to rectangular form, and identify the curve.

Multiply by $r$: $r^2 = 4r\cos\theta \Rightarrow x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0 \Rightarrow (x-2)^2 + y^2 = 4$. This is a circle of radius $2$ centered at $(2, 0)$.

## Identifying and sketching common polar curves

**3** Identify the type of curve for $r = 2 + 2\sin\theta$, and state its key feature.

This is a cardioid (since the coefficient of $\sin\theta$ equals the constant term) — a heart-shaped curve passing through the pole.

**4** Identify the type of curve for $r = \cos(3\theta)$.

This is a rose curve with $3$ petals (odd $n$ in $r = \cos(n\theta)$ gives $n$ petals).

## The slope of a polar curve

**5** State the formula for $\frac{dy}{dx}$ of a polar curve $r = f(\theta)$, using $x = r\cos\theta$ and $y = r\sin\theta$.

$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$.

**6** Find $\frac{dy}{dx}$ for $r = 2\theta$ at $\theta = \frac{\pi}{2}$.

$\frac{dr}{d\theta} = 2$. $x = 2\theta\cos\theta$, $y = 2\theta\sin\theta$. $\frac{dy}{d\theta} = 2\sin\theta + 2\theta\cos\theta$; $\frac{dx}{d\theta} = 2\cos\theta - 2\theta\sin\theta$. At $\theta = \frac{\pi}{2}$:
$\frac{dy}{d\theta} = 2(1) + 2\left(\frac{\pi}{2}\right)(0) = 2$; $\frac{dx}{d\theta} = 2(0) - 2\left(\frac{\pi}{2}\right)(1) = -\pi$. $\frac{dy}{dx} = \frac{2}{-\pi} = -\frac{2}{\pi}$.

## Finding horizontal and vertical tangents

**7** For the circle $r = 4\cos\theta$, find the value(s) of $\theta$ in $[0, \pi]$ where the tangent line is horizontal (where $y = r\sin\theta$ has a critical point).

$y = 4\cos\theta\sin\theta = 2\sin(2\theta)$. $\frac{dy}{d\theta} = 4\cos(2\theta) = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}$.

## Symmetry of polar curves

8 Explain how to test whether a polar curve $r = f(\theta)$ is symmetric about the polar axis (the $x$-axis).

Replace $\theta$ with $-\theta$ in the equation. If the equation is unchanged (or yields the same curve), the curve is symmetric about the polar axis. For example, $r = 2 + 2\cos\theta$ satisfies $f(-\theta) = 2 + 2\cos(-\theta) = 2 + 2\cos\theta = f(\theta)$, confirming symmetry.

9 Test whether $r = 3\sin\theta$ is symmetric about the $y$-axis (replace $\theta$ with $\pi - \theta$).

$f(\pi - \theta) = 3\sin(\pi - \theta) = 3\sin\theta = f(\theta)$ (using $\sin(\pi - \theta) = \sin\theta$). Symmetric about the $y$-axis.

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## More polar-to-rectangular conversions

10 Convert the rectangular point $(3, 3\sqrt{3})$ to polar coordinates ($r > 0$, $0 \leq \theta < 2\pi$).

$r = \sqrt{3^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = \sqrt{36} = 6$. $\theta = \tan^{-1}\left(\frac{3\sqrt{3}}{3}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$ (in Quadrant I, matching both coordinates positive).

11 Convert the rectangular equation $y = x$ to polar form.

$r\sin\theta = r\cos\theta \Rightarrow \tan\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$ (a constant-$\theta$ ray, i.e. a straight line through the pole).

12 Convert the polar equation $r\sin\theta = 4$ to rectangular form, and identify the curve.

$r\sin\theta = y$, so $y = 4$: a horizontal line.

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## More slope-of-a-polar-curve practice

13 Find $\frac{dy}{dx}$ for the cardioid $r = 1 + \sin\theta$ at $\theta = 0$.

$\frac{dr}{d\theta} = \cos\theta$. $\frac{dy}{dx} = \frac{\cos\theta\sin\theta + r\cos\theta}{\cos\theta\cos\theta - r\sin\theta}$. At $\theta = 0$: $r = 1$, $\frac{dr}{d\theta} = 1$. Numerator: $1(0) + 1(1) = 1$. Denominator: $1(1) - 1(0) = 1$. $\frac{dy}{dx} = \frac{1}{1} = 1$.

14 Find $\frac{dy}{dx}$ for the circle $r = 5$ (constant) at any angle $\theta$, and interpret the result using the fact that a circle centered at the origin has radial symmetry.

$\frac{dr}{d\theta} = 0$. $\frac{dy}{dx} = \frac{0 \cdot \sin\theta + 5\cos\theta}{0 \cdot \cos\theta - 5\sin\theta} = \frac{5\cos\theta}{-5\sin\theta} = -\cot\theta$. This matches the slope of the tangent line to a circle at angle $\theta$ — perpendicular to the radius at that point, consistent with basic circle geometry.

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## Tangent lines at the pole

15 Find the values of $\theta$ where the cardioid $r = 1 - \cos\theta$ passes through the pole ($r = 0$), and explain why these give the tangent line directions at the pole.

$1 - \cos\theta = 0 \Rightarrow \cos\theta = 1 \Rightarrow \theta = 0$. At the single angle where $r = 0$, the curve touches the pole; the tangent line there is along $\theta = 0$ (the polar axis), since that's the direction the curve approaches the origin from.

**16** Find the values of $\theta \in [0, 2\pi)$ where the rose $r = \cos(2\theta)$ passes through the pole.

$\cos(2\theta) = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ — these are the four tangent-line directions at the pole for this 4-petaled rose.

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## Classifying polar curves by equation type

**17** Classify each polar equation by curve type (circle, cardioid, limaçon with inner loop, rose, or line):

**a** $r = 5$ Circle centered at the pole

**b** $r = 3 + 3\cos\theta$ Cardioid (coefficient equals constant)

**c** $r = 1 + 3\sin\theta$ Limaçon with an inner loop (constant $<$ coefficient)

**d** $r = 4\sin(5\theta)$ Rose with $5$ petals (odd $n = 5$)

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## Number of petals in rose curves

**18** State the rule for the number of petals of $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$ depending on whether $n$ is odd or even.

If $n$ is odd, the rose has exactly $n$ petals (traced over $[0, \pi]$). If $n$ is even, the rose has $2n$ petals (requires the full $[0, 2\pi]$ to trace, since each petal is retraced with opposite sign after $\pi$).

**19** How many petals does $r = 2\sin(4\theta)$ have? How many does $r = 5\cos(7\theta)$ have?

$r = 2\sin(4\theta)$: $n = 4$ (even) $\Rightarrow 2(4) = 8$ petals. $r = 5\cos(7\theta)$: $n = 7$ (odd) $\Rightarrow 7$ petals.

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## Graphing polar curves point-by-point

**20** Complete a short table of $(\theta, r)$ values for $r = 2 + 2\cos\theta$ at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, and use it to describe the cardioid's shape.

$\theta = 0$: $r = 4$. $\theta = \frac{\pi}{2}$: $r = 2$. $\theta = \pi$: $r = 0$. $\theta = \frac{3\pi}{2}$: $r = 2$. The curve is widest ($r = 4$) along the polar axis, pinches to a point ($r = 0$) at $\theta = \pi$ (opposite direction), forming the characteristic heart shape.

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## Concavity considerations for polar curves

**21** Explain, in terms of $\frac{dy}{dx}$, what a 'vertical tangent' means for a polar curve, and why it corresponds to $\frac{dx}{d\theta} = 0$ (with $\frac{dy}{d\theta} \neq 0$).

A vertical tangent means the slope $\frac{dy}{dx}$ is undefined (infinite) — since $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$, this happens exactly when the denominator $\frac{dx}{d\theta} = 0$ while the numerator stays nonzero, making the ratio blow up.